

Sommatoria	Somma parziale	Complessità
$S = \sum_{i=1}^n i$	$S = \frac{n(n+1)}{2}$	$S = \Theta(n^2)$
$S = \sum_{i=0}^n c^i$	$S = \frac{c^{n+1}-1}{c-1}$	$S = \Theta(c^n)$
$\sum_{i=0}^x c^i = \Theta(1)$ quando $c < 1$		
$S = \sum_{i=1}^n i \cdot c^i$	$S = \frac{nc^{n+1}}{c-1} - \frac{c^{n+1}-1}{(c-1)^2} + 1$	$S = \Theta(nc^n)$
$S = \sum_{i=1}^n \log_2^c i$		$S = \Theta(n \log^c n)$
$S = \sum_{i=1}^n \frac{1}{i}$		$S = \Theta(\log n)$
$S = \sum_{i=1}^n i^c$		$S = \Theta(n^{c+1})$

$a^{\log_a(b)} = b$ con $a > 0$ e $a \neq 1$ e $b > 0$	$\log_a(bc) = \log_a(b) + \log_a(c)$
$\log_a(b^c) = c \log_a(b)$	$\log_a(\frac{b}{c}) = \log_a(b) - \log_a(c)$
$\log_a(b) = \frac{\log_c(b)}{\log_c(a)}$ con $c > 1$	$\log_a(b) = \frac{1}{\log_b(a)}$

Sommatoria	Risultato
$\sum_{i=0}^n i^c$	$\Theta(n^{c+1})$
$\sum_{i=0}^n c^i$	$\frac{c^{n+1}-1}{c-1}$ per ogni c positivo diverso da 1 per cui: $S = \Theta(c^n)$ se $c > 1$ e $S = O(1)$ se $c < 1$
$\sum_{i=0}^n i c^i$	$\Theta(n c^n)$ per $c > 1$
$\sum_{i=0}^n \log^c i$	$\Theta(n \log^c n)$ per ogni $c > 1$
$\sum_{i=0}^n \frac{1}{i}$	$\Theta(\log n)$